ECON 1550 — Problem Set 9 Answer Key

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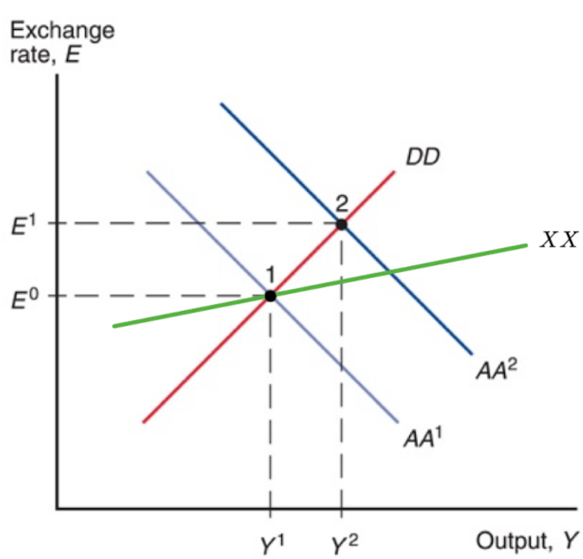
Submission:	Canvas or Gradescope

1. **Devaluation and the Current Account**

Assume the economy is at its long-run equilibrium and that the current account is in balance (equal to zero). The central bank is implementing a fully credible fixed exchange rate regime. Using the AA-DD model, explain why a devaluation improves the current account in the short run.

Solution: As shown in the figure below, a devaluation causes the AA curve to shift from AA^1 to AA^2, which reflects an expansion in the money supply needed to sustain the new peg. The figure also shows an XX curve in green along which the current account is in balance. The initial equilibrium, at point 1, was on the XX curve, reflecting the fact that the current account was in balance there. After the devaluation, the new equilibrium point is above the XX curve, in the region where the current account is in surplus. With fixed prices, a devaluation improves an economy's competitiveness, increasing its exports, decreasing its imports, and raising the level of output.



2. **A Flat AA Curve**

Consider the following version of the AA-DD model:

Consumption	:	$C = (1-a)Y$
Investment	:	I
Government spending	:	G
Current account	:	$CA = b\dfrac{EP^*}{P} - cY$
Demand for domestic goods	:	$D = C + I + G + CA$
Real money demand	:	$\dfrac{M^d}{P} = 1 + d\left(Y - Y^f\right) - eR$
Real money supply	:	$\dfrac{M^s}{P}$
Uncovered interest parity	:	$R = R^* + \dfrac{E^e}{E} - 1$

The variables P^*, I, G, Y^f, M^s, and R^* are exogenous. The parameters a, b, c, d, and e are also exogenous and assumed to be positive. The variables C, Y, CA, E, P, D, E^e, M^d, and R are endogenous.

Assume that:

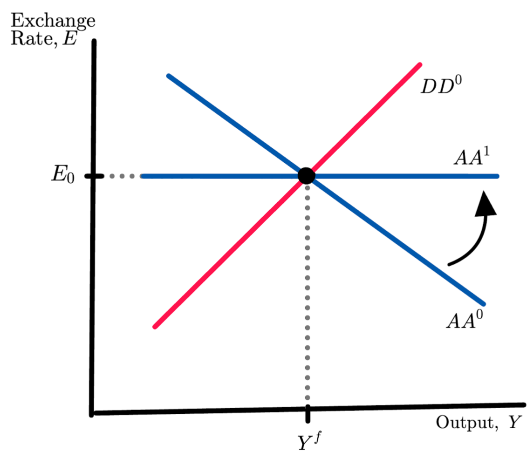
- in the long run, $Y = Y^f$, $E = E^e$ and P is fully flexible;
- in the short run, P is fixed, and E^e is equal to the value of E in the final long-run equilibrium;
- the economy is initially in a long-run equilibrium with output equal to the full-employment level Y^f.

(a) Give intuition for why the real money demand is increasing in Y and identify the parameter that determines how real money demand responds to Y.

Solution: When domestic income Y increases, people demand more money to be able to carry out more transactions, because higher income raises demand for goods and services. The parameter that determines how real money demand responds to Y is d.

(b) Show in an AA-DD diagram the effects of a permanent change in the parameter d from its initial positive value to a new value of $d = 0$, and state whether the equilibrium (Y, E) changes.

Solution: The DD equation does not change since d only shows up in the real money demand equation. The AA curve becomes flat but still goes through the same initial equilibrium:



Therefore, even though d changed, the economy remained in the same equilibrium.

<u>Explanation (not part of the answer, not needed for credit)</u>

The reason for the AA becoming flat is that real money demand becomes unresponsive to Y. In the derivation of the AA curve, when we considered different values of Y, there was an eventual change in E because real money demand changes with Y. If real money demand does not change with Y, the equilibrium in the money market is the same for all Y, which then implies, through the interest parity condition, that the equilibrium in the foreign exchange market remains unchanged as well.

Since the economy with $d > 0$ is at an equilibrium with $Y = Y^f$, the economy with $d = 0$ is also at the same equilibrium since in both cases real money demand is the same.

(c) For this question, assume that the economy starts at the long-run equilibrium that is reached after d changes to zero permanently, and continue to assume that $d = 0$ permanently. Consider a temporary fiscal expansion that increases G to a higher value G^{high}. In the short run, does the current account increase, decrease, or stay the same? Give economic intuition for your answer.

Solution: The current account deteriorates.
The economic intuition is that the fiscal expansion increases aggregate demand, which shifts the DD to the right. Higher aggregate demand translates into higher output and hence higher income. With higher income, consumers want to consume more domestic and more foreign goods. The desire to consume more foreign goods means import demand increases which, in equilibrium, increases imports. Exports don't change since foreign income is unchanged and the exchange rate remains fixed because the AA is flat. Higher imports and unchanged exports imply a deterioration in the current account.
Explanation (not part of the answer, not needed for credit)
A fast way to realize that the current account deteriorates is to use the XX curve that goes through the initial equilibrium. After the shift in the DD, the new equilibrium is below the XX curve, so the current account deteriorates.

(d) Now assume that, before the temporary fiscal expansion, the central bank implements a fixed exchange rate regime at the prevailing exchange rate, which is the long-run level with $d = 0$. Are the short-run effects of fiscal expansion on the current account the same as in the last question? Why or why not?

Solution: A flat AA effectively fixes the exchange rate with or without a fixed exchange rate regime. Thus, the effects of the fiscal expansion are the same.
Explanation (not part of the answer, not needed for credit)
This problem shows that if the real money demand does not respond to income/output, then the economy is, effectively, in a fixed exchange rate regime. This means that when the real money demand *does* depend on income/output, all the central bank must do to keep the exchange rate fixed is to undo the effects that changes in output have on the real money demand.

3. **A Preview of the II-XX Model**

This question introduces the II-XX model, a framework for analyzing internal and external balance in an open economy. Everything you need is given below; you do not need prior knowledge of this model. For optional background, you may consult the lecture notes chapter on the II-XX model, but it is not required to answer any part of this question.

The current account is given by:

$$\text{CA}(q, Y) = d_1 q - d_2 Y, \tag{1}$$

where q is the real exchange rate, Y is output, and d_1, d_2 are positive parameters. Domestic demand (absorption) is

$$A = C + G, \tag{2}$$

where G is government spending, and C is consumption, which is given by

$$C = c_0 + c_1 Y, \tag{3}$$

where c_0 is a positive parameter and $c_1 \in (0, 1)$ is the marginal propensity to consume. We denote the full-employment level of output by Y^f and treat it as exogenous.

The II curve in this economy is given by pairs A, q that make the goods-market be in equilibrium when output is at full employment:

$$Y^f = A + \text{CA}\left(q, Y^f\right). \tag{4}$$

The XX curve is given by pairs A, q that keep the current account constant at its long-run sustainable level XX^b:

$$XX^b = \text{CA}(q, A), \tag{5}$$

where XX^b is an exogenous constant level for the current account, and $\text{CA}(q, A) = \text{CA}(q, Y)$ is the current account expressed as a function of q and A instead of q and Y.

The economy is in "internal balance" when $Y = Y^f$ and in "external balance" when $CA = XX^b$. Both are stable rest points: absent shocks, Y converges to Y^f and CA converges to XX^b, and both remain there once reached.

(a) Assume that, for imports, the volume effect dominates the value effect. Give economic intuition for why CA is increasing in q and decreasing in Y.

Solution: Intuition for CA increasing in q

When there is a real depreciation, foreign goods and services become more expensive relative to domestic goods and services. Foreign consumers demand more of the domestic country's goods and services, which increases exports EX. Domestic consumers demand fewer of the foreign country's goods and services, which reduces imports IM, assuming the volume effect dominates the value effect.

Intuition for CA decreasing in Y

When Y increases, domestic income is higher. Domestic consumers increase their spending on all goods, both domestic and foreign. The higher spending on foreign goods makes IM go up. Exports EX remain unchanged because they do not depend on domestic demand. Since $CA = EX - IM$, unchanged EX and higher IM lead to a lower CA.

(b) Use equations (2) and (3) to express Y as a function of A. Then find the function $\text{CA}(q, A)$ explicitly.

Solution: Substituting (3) into (2):

$$A = c_0 + c_1 Y + G.$$

Solving for Y:

$$Y = \frac{1}{c_1} A - \frac{c_0 + G}{c_1}. \tag{6}$$

Substituting (6) into (1) gives

$$\text{CA}(q, A) = d_1 q - d_2 \left(\frac{1}{c_1} A - \frac{c_0 + G}{c_1} \right) = d_1 q - \frac{d_2}{c_1} A + \frac{d_2(c_0 + G)}{c_1}.$$

(c) Find the function $f(\cdot)$ such that equation (4) can be written as $q = f(A)$. Is f increasing, decreasing, or constant? Give economic intuition.

Solution: Substituting (1) into (4):

$$Y^f = A + d_1 q - d_2 Y^f.$$

Solving for q:

$$q = -\frac{1}{d_1}A + \frac{(1+d_2)Y^f}{d_1}.$$

Therefore,

$$f(A) = -\frac{1}{d_1}A + \frac{(1+d_2)Y^f}{d_1}.$$

The function f is decreasing.

Intuition: when domestic demand A increases, the level of the current account needed to keep $Y = Y^f$ goes down. A lower CA consistent with Y^f can only be achieved with a real appreciation, that is, a lower q.

(d) Find the function $g(\cdot)$ such that equation (5) can be written as $q = g(A)$. Is g increasing, decreasing, or constant? Give economic intuition.

Solution: Using equations (1) and (5):

$$XX^b = \text{CA}(q, A) = \text{CA}(q, Y) = d_1 q - d_2 Y.$$

Substituting (6):

$$XX^b = d_1 q - d_2\left(\frac{1}{c_1}A - \frac{c_0 + G}{c_1}\right). \tag{7}$$

Solving for q:

$$q = \frac{d_2}{c_1 d_1}A + \frac{1}{d_1}XX^b - \frac{d_2 G}{c_1 d_1} - \frac{c_0 d_2}{c_1 d_1}.$$

Therefore,

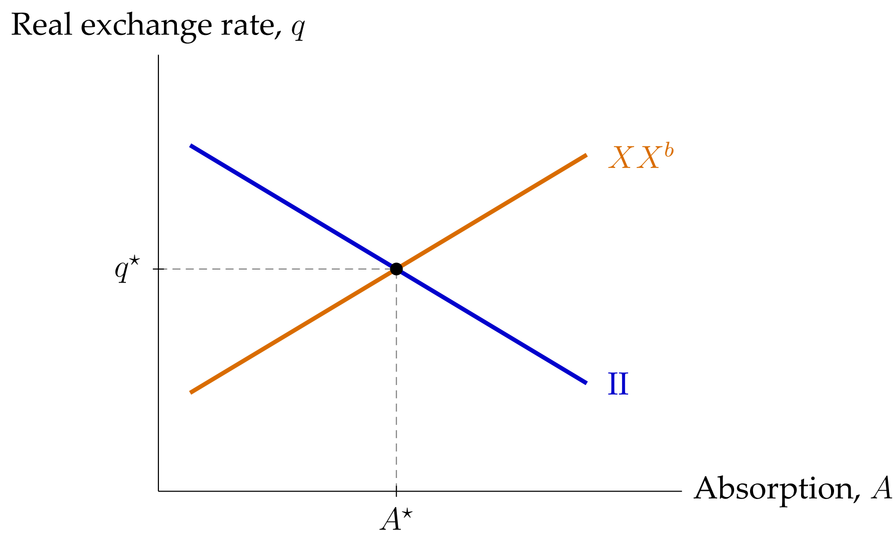
$$g(A) = \frac{d_2}{c_1 d_1}A + \frac{1}{d_1}XX^b - \frac{d_2 G}{c_1 d_1} - \frac{c_0 d_2}{c_1 d_1}.$$

The function g is increasing.

Intuition: when domestic demand A increases, demand for both domestic and foreign goods and services increases. Higher demand for foreign goods raises imports and thus deteriorates CA. To keep CA at the fixed level XX^b while supporting a higher level of A, the real exchange rate must depreciate (q must rise): the real depreciation raises exports and lowers imports, offsetting the import-driven deterioration.

(e) Sketch the II and XX^b curves in a diagram with A on the horizontal axis and q on the vertical axis. Label the intersection as $(A^\star, q^\star)$.

Solution:



(f) Assume the economy is initially at the intersection of the II and XX^b curves. The government permanently increases G. Do the II and XX^b curves shift? Explain why, using your expression for $\text{CA}(q, A)$ from (b).

Solution: The II curve does not shift

From equation (4), the II curve is defined by $Y^f = A + \text{CA}(q, Y^f)$. For a given level of domestic demand A, the level of CA required to keep $Y = Y^f$ depends only on Y^f and on the function $\text{CA}(\cdot, \cdot)$ in equation (1), neither of which depends on G. Hence the q consistent with $Y = Y^f$ at a given A does not change. Concretely, the expression for $f(A)$ derived in (c) contains no G:

$$f(A) = -\frac{1}{d_1}A + \frac{(1 + d_2)Y^f}{d_1}.$$

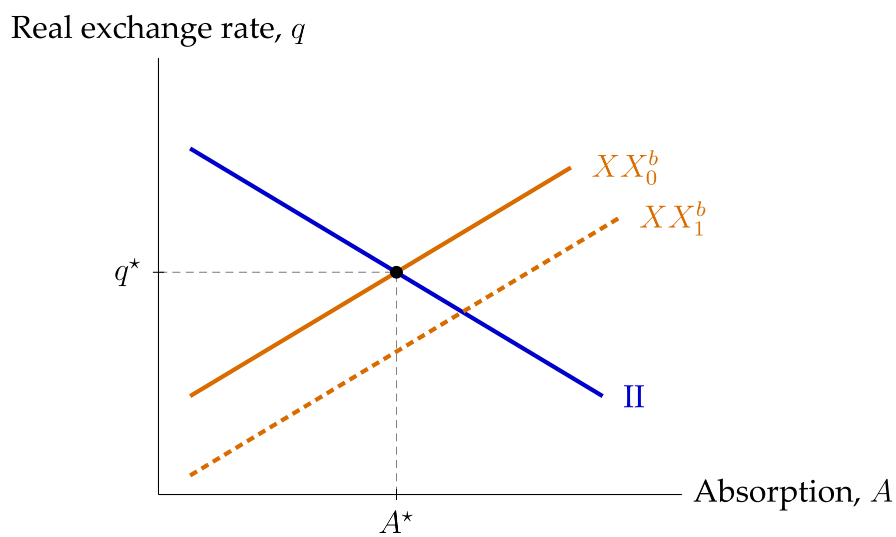
Note that while G does affect A through equations (2)–(3), the higher G induces a movement along the II curve rather than a shift in the curve.

The XX^b curve shifts down

Pick a fixed value for A, say $A = \bar{A}$. With A fixed at $\bar{A}$, higher G implies lower C via equation (2) (to keep A constant, the higher G must be offset by a lower C). Lower C requires lower Y via equation (3). Lower Y reduces imports (since CA is decreasing in Y), which improves CA. To keep $CA = XX^b$, this import-driven improvement must be offset by a real appreciation, meaning a lower q, which worsens CA through the exports-minus-imports channel. Therefore at the fixed $A = \bar{A}$, the value of q consistent with external balance is lower after the increase in G: the XX^b curve shifts down.

Explanation (not part of the answer, not needed for credit)

Equation (7) shows the shift algebraically. The function $g(A)$ derived in (d) contains a $-d_2 G/(c_1 d_1)$ term; increasing G shifts the whole function down by that amount.



(g) Immediately after the increase in G, assume q and Y are unchanged on impact, so $q = q^\star$ and $Y = Y^f$. What is the new value of A? Is the economy in internal balance, external balance, both, or neither? Explain.

Solution: New value of A

Let $\Delta G > 0$ denote the increase in G. On impact, with Y held at Y^f, consumption $C = c_0 + c_1 Y^f$ is unchanged. From equation (2), absorption then rises one-for-one with G:

$$A_{\text{new}} = A^\star + \Delta G.$$

External balance holds

The current account $\text{CA}(q, Y) = d_1 q - d_2 Y$ depends on q and Y only, not directly on G. With $q = q^\star$ and $Y = Y^f$ unchanged on impact, CA is unchanged at its initial level:

$$\text{CA}(q^\star, Y^f) = d_1 q^\star - d_2 Y^f = XX^b.$$

So external balance is maintained. Equivalently, in the (A, q) diagram the XX^b curve shifted down by $d_2\Delta G/(c_1 d_1)$ in (f), and A moves to the right by ΔG; since g has slope $d_2/(c_1 d_1)$, the vertical drop of the curve and the move along the curve exactly cancel, so the point $(A_{\text{new}}, q^\star)$ lies on the new XX_1^b curve.

<u>Internal balance does not hold</u>

The II curve requires $Y^f = A + \text{CA}(q, Y^f)$. At $(A_{\text{new}}, q^\star)$,

$$A_{\text{new}} + \text{CA}(q^\star, Y^f) = (A^\star + \Delta G) + XX^b = Y^f + \Delta G > Y^f.$$

Goods demand at full employment exceeds Y^f, so the point $(A_{\text{new}}, q^\star)$ is above the II curve and internal balance fails. Intuitively, at the new A and unchanged q, holding $Y = Y^f$ leaves excess demand for goods: the goods-market equilibrium level of output would be above Y^f.

Therefore, on impact the economy is in external balance but not in internal balance.